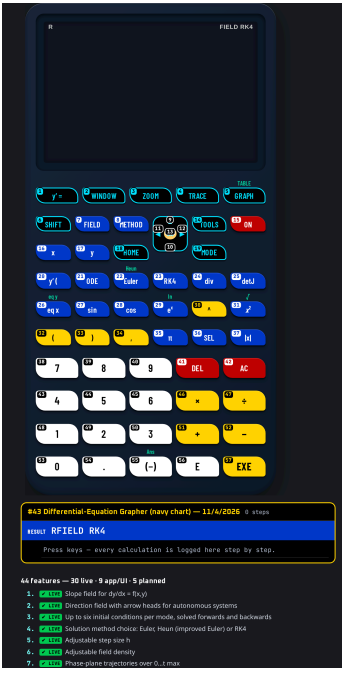
U-45 CALCULATORS

User Manual

Calculator #43 Differential-Equation Grapher (navy chart)

Graphing – colour



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#43 — Differential-Equation Grapher (navy chart)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example 

. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

differential-equation graphing: slope fields, Euler/Heun/RK4 solutions and phase planes.

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
10. Practice drills
11. Appendix A — Drill answers

12. Appendix B — Function finder (A–Z)
13. Appendix C — Glossary
14. Appendix D — Model notes & limits
15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#43** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#38, #39, #40, #43, #44, #45, #48, #49, #50, #53, #54, #55, #56
op	Operators & execute	#13, #30, #32, #33, #34, #46, #47, #51, #52, #57
fn	Functions	#7, #8, #16, #17, #20, #21, #22, #23, #24, #25, #26, #27, #28, #29, #31, #35, #36, #37
mem	Memory, variables & registers	—
clr	Clear, delete & power	#15, #41, #42

mod	**Shift, modes, menus & navigation**	**#1, #2, #3, #4, #5, #6, #9, #10, #11, #12, #14, #18, #19**

2 · Features

44 features: 30 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE Slope field for dy/dx = f(x,y)
2. LIVE Direction field with arrow heads for autonomous systems
3. LIVE Up to six initial conditions per mode, solved forwards and backwards
4. LIVE Solution method choice: Euler, Heun (improved Euler) or RK4
5. LIVE Adjustable step size h
6. LIVE Adjustable field density
7. LIVE Phase-plane trajectories over 0…t max
8. LIVE Nullclines (x′ = 0 and y′ = 0) or the zero-slope isocline
9. LIVE Isocline y′ = c tool
10. LIVE Trace along a solution with y and slope read-out
11. LIVE Trace phase trajectories in time steps
12. LIVE Switch traced curve with ▲ ▼
13. LIVE Tool: solve y(x) from any initial condition
14. LIVE Tool: Euler / Heun / RK4 comparison with error against a fine reference
15. LIVE Comparison curves drawn dashed
16. LIVE Tool: equilibrium point by Newton's method
17. LIVE Equilibrium classification (node, spiral, saddle, centre) from the Jacobian
18. LIVE Tool: add or clear initial conditions
19. LIVE FIELD key toggles slope field / phase plane

20. LIVE METHOD key cycles the integrator
21. LIVE Table comparing Euler, Heun and RK4 from the first IC
22. LIVE Home slope(x,y)
23. LIVE Home ode(x_0,y_0,x) high-accuracy RK4 (h = 0.01)
24. LIVE Home euler(), heun(), rk4() with n steps
25. LIVE Home divergence divg(x,y) of the system
26. LIVE Home Jacobian determinant jdet(x,y)
27. LIVE Home equilibrium coordinates eqx(), eqy()
28. LIVE Damped double-well (Duffing-type) oscillator preset
29. LIVE Dark chart display with colour-coded curves
30. LIVE Window form with step, t max and density
31. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
32. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
33. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
34. APP/UI Show / hide key numbers with one button (remembered between sessions)
35. APP/UI Copy the current display text to the clipboard
36. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
37. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
38. APP/UI Reference photo viewer and tool-tip name on every key

39. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
40. PLANNED **Graph image export (PNG)**
41. PLANNED **Shareable function list link**
42. PLANNED **Dark screen theme**
43. PLANNED **Speech read-out of values**
44. PLANNED **Exam mode**

2b · Feature guide

Calculator #43 is a differential-equation graphing: slope fields, Euler/Heun/RK4 solutions and phase planes. Each card below is one of its 30 working features, with the keys that reach it and, where one exists, a verified example.

F1 Slope field for dy/dx = f(x,y)

Where: #7 FIELD Toggle slope field / phase plane; #20 y′ (Slope field value

See it in action: Slope of y′ = x − y at (2, 1) — *slope(2,1)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Direction field with arrow heads for autonomous systems

Where: #7 FIELD Toggle slope field / phase plane; #20 y′ (Slope field value

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

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F3 Up to six initial conditions per mode, solved forwards and backwards

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Solution method choice: Euler, Heun (improved Euler) or RK4

Where: #8 METHOD Euler → Heun → RK4; #22 Euler Euler's method with n steps; #21 ODE Accurate RK4 solution

See it in action: RK4 solution of y′ = x − y, y(0)=1, at x = 1 (exact 2/e) — *ode(0,1,1)*= 0.73575888; Euler's method, 10 steps, same problem — *euler(0,1,1,10)*= 0.69735688

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Adjustable step size h

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Adjustable field density

Where: #7 FIELD Toggle slope field / phase plane; #20 y' (Slope field value

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Phase-plane trajectories over 0…t max

Where: #7 FIELD Toggle slope field / phase plane

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Nullclines (x′ = 0 and y′ = 0) or the zero-slope isocline

Where: #7 FIELD Toggle slope field / phase plane; #20 y′(Slope field value

See it in action: Slope of y′ = x − y at (2, 1) — *slope(2,1)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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F9 Isocline y′ = c tool

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Trace along a solution with y and slope read-out

Where: #4 TRACE TRACE — move along a graph; #7 FIELD Toggle slope field / phase plane; #20 y' (Slope field value

See it in action: Slope of y′ = x − y at (2, 1) — *slope(2,1)*= 1; RK4 solution of y′ = x − y, y(0)=1, at x = 1 (exact 2/e) — *ode(0,1,1)*= 0.73575888

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Trace phase trajectories in time steps

Where: #4 TRACE TRACE — move along a graph; #7 FIELD Toggle slope field / phase plane; #22 Euler Euler's method with n steps

See it in action: Euler's method, 10 steps, same problem — *euler(0,1,1,10)*= 0.69735688

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Switch traced curve with ▲ ▼

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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F13 Tool: solve y(x) from any initial condition

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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F14 Tool: Euler / Heun / RK4 comparison with error against a fine reference

Where: #8 METHOD Euler → Heun → RK4; #22 Euler Euler's method with n steps; #21 ODE Accurate RK4 solution

See it in action: RK4 solution of y′ = x − y, y(0)=1, at x = 1 (exact 2/e) — *ode(0,1,1)*= 0.73575888; Euler's method, 10 steps, same problem — *euler(0,1,1,10)*= 0.69735688

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Comparison curves drawn dashed

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

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F16 Tool: equilibrium point by Newton's method

Where: #8 METHOD Euler → Heun → RK4; #22 Euler Euler's method with n steps; #26 eq x Equilibrium x from a guess

See it in action: Euler's method, 10 steps, same problem — *euler(0,1,1,10)* = 0.69735688; Jacobian determinant at the equilibrium (1, 0) — *jdet(1,0)* = 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Equilibrium classification (node, spiral, saddle, centre) from the Jacobian

Where: #25 detJ Jacobian determinant; #26 eq x Equilibrium x from a guess

See it in action: Jacobian determinant at the equilibrium (1, 0) — *jdet(1,0)* = 2; Equilibrium found from guess (0.8, 0.1): x-coordinate — *eqx(0.8,0.1)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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F18 Tool: add or clear initial conditions

Where: #42 AC **AC — clear all of the current calculation**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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_____

F19 FIELD key toggles slope field / phase plane

Where: #7 FIELD **Toggle slope field / phase plane;** #20 y' (**Slope field value**

See it in action: Slope of y′ = x − y at (2, 1) — *slope(2,1)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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F20 METHOD key cycles the integrator

Where: #8 METHOD Euler → Heun → RK4; #22 Euler Euler's method with n steps

See it in action: Euler's method, 10 steps, same problem — *euler(0,1,1,10)*= 0.69735688

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 Table comparing Euler, Heun and RK4 from the first IC

Where: #8 METHOD Euler → Heun → RK4; #22 Euler Euler's method with n steps; #5 GRAPH GRAPH — draw the graph

See it in action: RK4 solution of y′ = x − y, y(0)=1, at x = 1 (exact 2/e) — *ode(0,1,1)*= 0.73575888; Euler's method, 10 steps, same problem — *euler(0,1,1,10)*= 0.69735688

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Home slope(x,y)

Where: #7 FIELD Toggle slope field / phase plane; #20 y' (Slope field value

See it in action: Slope of y′ = x − y at (2, 1) — *slope(2,1)*= 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Home ode(x_0,y_0,x) high-accuracy RK4 (h = 0.01)

Where: #21 ODE Accurate RK4 solution; #8 METHOD Euler → Heun → RK4; #23 RK4 Runge-Kutta 4 with n steps

See it in action: RK4 solution of y′ = x − y, y(0)=1, at x = 1 (exact 2/e) — *ode(0,1,1)*= 0.73575888

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 Home euler(), heun(), rk4() with n steps

Where: #8 METHOD Euler → Heun → RK4; #22 Euler Euler's method with n steps; #23 RK4 Runge-Kutta 4 with n steps

See it in action: RK4 solution of $y' = x - y$, $y(0)=1$, at $x = 1$ (exact $2/e$) — *ode(0,1,1)*= 0.73575888; Euler's method, 10 steps, same problem — *euler(0,1,1,10)*= 0.69735688

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F25 Home divergence divg(x,y) of the system

Where: #24 div Divergence of the system; #1 y' = Edit the equation and system

See it in action: Divergence of the system $x' = y$, $y' = x - x^3 - 0.3y$ — *divg(0,0)*= -0.3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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_____

F26 Home Jacobian determinant jdet(x,y)

Where: #25 detJ Jacobian determinant

See it in action: Jacobian determinant at the equilibrium (1, 0) — *jdet(1,0)* = 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Home equilibrium coordinates eqx(), eqy()

Where: #26 eq x Equilibrium x from a guess; #34 , Comma / argument separator (between function arguments, list items or coordinates)

See it in action: Jacobian determinant at the equilibrium (1, 0) — *jdet(1,0)* = 2; Equilibrium found from guess (0.8, 0.1): x-coordinate — *eqx(0.8,0.1)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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F28 Damped double-well (Duffing-type) oscillator preset

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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_____

F29 Dark chart display with colour-coded curves

Where: #19 MODE SETUP — angle unit, number format, display options

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

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F30 Window form with step, t max and density

Where: #2 WINDOW WINDOW — set the graph view

Try it: Press Reset, use the key(s) above, and write the display here:

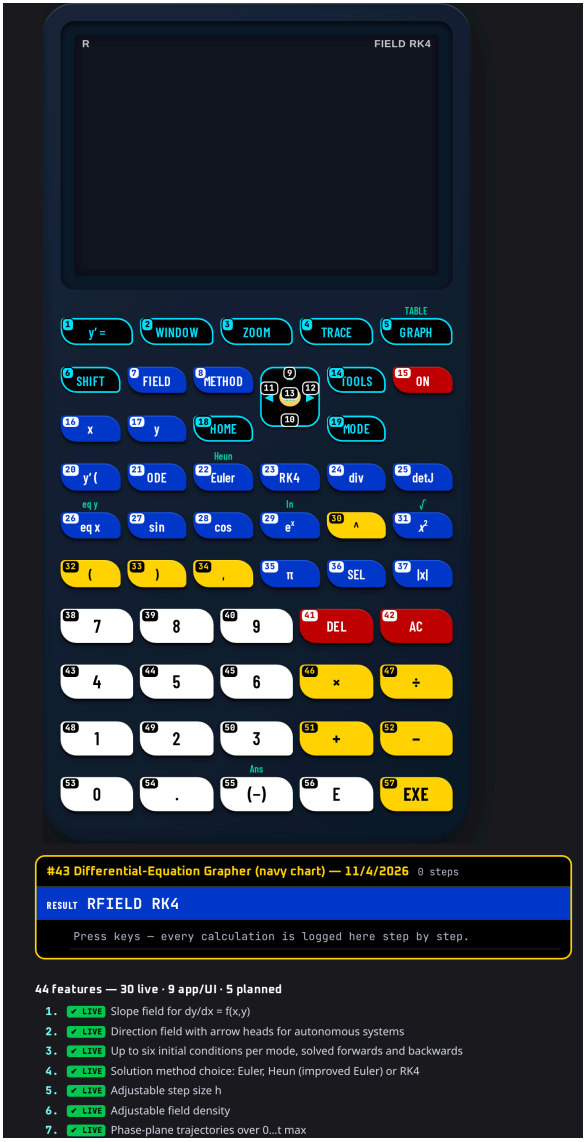
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #43 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	y′ =	·	·	·	mod
2	WINDOW	·	·	·	mod
3	ZOOM	·	·	·	mod
4	TRACE	·	·	·	mod
5	GRAPH	TABLE	·	·	mod
6	SHIFT	·	·	·	mod
7	FIELD	·	·	·	fn
8	METHOD	·	·	·	fn
9	▲	·	·	·	mod
10	▼	·	·	·	mod
11	◀	·	·	·	mod
12	▶	·	·	·	mod
13	OK	·	·	·	op
14	TOOLS	·	·	·	mod
15	ON	·	·	·	clr
16	x	·	·	·	fn
17	y	·	·	·	fn
18	HOME	·	·	·	mod
19	MODE	·	·	·	mod
20	y′ (	·	·	·	fn
21	ODE	·	·	·	fn
22	Euler	Heun	·	·	fn
23	RK4	·	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	div	·	·	·	fn
25	detJ	·	·	·	fn
26	eq x	eq y	·	·	fn
27	sin	·	·	·	fn
28	cos	·	·	·	fn
29	ex	ln	·	·	fn
30	^	·	·	·	op
31	x^2	√	·	·	fn
32	(	·	·	·	op
33	)	·	·	·	op
34	,	·	·	·	op
35	π	·	·	·	fn
36	SEL	·	·	·	fn
37	|x|	·	·	·	fn
38	7	·	·	·	num
39	8	·	·	·	num
40	9	·	·	·	num
41	DEL	·	·	·	clr
42	AC	·	·	·	clr
43	4	·	·	·	num
44	5	·	·	·	num
45	6	·	·	·	num
46	×	·	·	·	op
47	÷	·	·	·	op
48	1	·	·	·	num
49	2	·	·	·	num
50	3	·	·	·	num
51	+	·	·	·	op

#	Face	Shift	Alpha	3rd	Group
52	−	·	·	·	op
53	0	·	·	·	num
54	.	·	·	·	num
55	(−)	Ans	·	·	num
56	E	·	·	·	num
57	EXE	·	·	·	op

4 · Key-by-key reference

Every key on calculator #43, in number order, with every function it carries in this app.

#1 y′ = `Shift, modes, menus & navigation`

Primary: Edit the equation and system

#2 WINDOW `Shift, modes, menus & navigation`

Primary: WINDOW — set the graph view

#3 ZOOM `Shift, modes, menus & navigation`

Primary: ZOOM — zoom presets

#4 TRACE `Shift, modes, menus & navigation`

Primary: TRACE — move along a graph

#5 GRAPH `Shift, modes, menus & navigation`

Primary: GRAPH — draw the graph

shifted (2nd / f / SHIFT layer) (press #6 first): TABLE — TABLE — table of values

#6 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#7 FIELD Functions

Primary: Toggle slope field / phase plane

#8 METHOD Functions

Primary: Euler → Heun → RK4

#9 ▲ Shift, modes, menus & navigation

Primary: Cursor ▲ — up / previous history line / previous item

#10 ▼ Shift, modes, menus & navigation

Primary: Cursor ▼ — down / next history line / next item

#11 ◀ Shift, modes, menus & navigation

Primary: Cursor ◀ — move left in the line

#12 ▶ Shift, modes, menus & navigation

Primary: Cursor ▶ — move right in the line

#13 OK Operators & execute

Primary: OK — confirm the highlighted item

#14 TOOLS Shift, modes, menus & navigation

Primary: TOOLS — tools menu

#15 ON Clear, delete & power

Primary: ON — power on

#16 x Functions

Primary: Variable x

#17 y Functions

Primary: Variable y

#18 HOME Shift, modes, menus & navigation

Primary: HOME — return to the main calculation screen

#19 MODE Shift, modes, menus & navigation

Primary: SETUP — angle unit, number format, display options

#20 y' (Functions

Primary: Slope field value

#21 ODE Functions

Primary: Accurate RK4 solution

#22 Euler Functions

Primary: Euler's method with n steps

shifted (2nd / f / SHIFT layer) (press #6 first): Heun — Heun's method

#23 RK4 Functions

Primary: Runge-Kutta 4 with n steps

#24 div Functions

Primary: Divergence of the system

#25 detJ Functions

Primary: Jacobian determinant

#26 eq x Functions

Primary: Equilibrium x from a guess

shifted (2nd / f / SHIFT layer) (press #6 first): eq y — Equilibrium y

#27 sin Functions

Primary: Sine (uses the current angle unit)

Keyboard: s

#28 cos Functions

Primary: Cosine (current angle unit)

Keyboard: c

#29 ex Functions

Primary: e^x — exponential

shifted (2nd / f / SHIFT layer) (press #6 first): ln — Natural logarithm ln

#30 ^ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

Keyboard: ^

#31 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #6 first): √ — Square root √

Keyboard: q

#32 (Operators & execute

Primary: Opening bracket (

Keyboard: (

#33) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#34 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#35 π Functions

Primary: π (3.14159265…) as an exact constant

Keyboard: p

#36 SEL Functions

Primary: sel

#37 |x| Functions

Primary: Absolute value |x|

#38 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#39 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#40 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#41 DEL Clear, delete & power

Primary: DEL — delete the character at the cursor

Keyboard: Backspace Delete

#42 AC Clear, delete & power

Primary: AC — clear all of the current calculation

Keyboard: Escape

#43 4 Digits & number entry

Primary: Digit 4

Keyboard: 4

#44 **5** Digits & number entry

Primary: Digit 5

Keyboard: 5

#45 **6** Digits & number entry

Primary: Digit 6

Keyboard: 6

#46 **×** Operators & execute

Primary: Multiplication ×

Keyboard: * x

#47 **÷** Operators & execute

Primary: Division ÷

Keyboard: /

#48 **1** Digits & number entry

Primary: Digit 1

Keyboard: 1

#49 **2** Digits & number entry

Primary: Digit 2

Keyboard: 2

#50 3 Digits & number entry

Primary: Digit 3

Keyboard: 3

#51 + Operators & execute

Primary: Addition +

Keyboard: +

#52 – Operators & execute

Primary: Subtraction –

Keyboard: -

#53 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#54 . Digits & number entry

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

Keyboard: .

#55 (−) Digits & number entry

Primary: Negative sign (−) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #6 first): Ans — Ans — recall the last answer into the current expression

#56 E Digits & number entry

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: e

#57 EXE Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: Enter =

5 · Worked examples

6 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Slope of y′ = x − y at (2, 1)

Calculate **slope(2,1)**. Press **Reset** first.

`#20` y′ (→ `#49` 2 → `#34` , → `#48` 1 → `#33`) → `#57` EXE

Display shows `1` — expected 1. ✔ verified on the app.

Example 5.2 — RK4 solution of y′ = x − y, y(0)=1, at x = 1 (exact 2/e)

Calculate **ode(0,1,1)**. Press **Reset** first.

`#21` ODE → `#53` 0 → `#34` , → `#48` 1 → `#34` , → `#48` 1 → `#33`) → `#57` EXE

Display shows `0.7357588824` — expected 0.73575888. ✔ verified on the app.

Example 5.3 — Euler's method, 10 steps, same problem

Calculate **euler(0,1,1,10)**. Press **Reset** first.

`#22` Euler → `#53` 0 → `#34` , → `#48` 1 → `#34` , → `#48` 1 → `#34` , → `#48` 1 → `#53` 0 → `#33`) → `#57` EXE

Display shows `0.6973568802` — expected 0.69735688. ✔ verified on the app.

Example 5.4 — Divergence of the system $x' = y, y' = x - x^3 - 0.3y$

Calculate **divg(0,0)**. Press **Reset** first.

#24 div → #53 0 → #34 , → #53 0 → #33) → #57 EXE

Display shows `-0.3` — expected -0.3. ✔ verified on the app.

Example 5.5 — Jacobian determinant at the equilibrium (1, 0)

Calculate **jdet(1,0)**. Press **Reset** first.

#25 detJ → #48 1 → #34 , → #53 0 → #33) → #57 EXE

Display shows `2` — expected 2. ✔ verified on the app.

Example 5.6 — Equilibrium found from guess (0.8, 0.1): x-coordinate

Calculate **eqx(0.8,0.1)**. Press **Reset** first.

#26 eq x → #53 0 → #54 . → #39 8 → #34 , → #53 0 → #54 . → #48 1 → #33) → #57 EXE

Display shows `1` — expected 1. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

6 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #5 → TABLE; #6 then #22 → Heun; #6 then #26 → eq y; #6 then #29 → ln.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #57.

Hack 18: Type faster on a computer

27 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
s	#27	sin	Sine (uses the current angle unit)
c	#28	cos	Cosine (current angle unit)
^	#30	^	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
q	#31	x^2	Square x^2
(	#32	(	Opening bracket (
)	#33	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#34	,	Comma / argument separator (between function arguments, list items or coordinates)
p	#35	π	π (3.14159265…) as an exact constant
7	#38	7	Digit 7
8	#39	8	Digit 8
9	#40	9	Digit 9
Backspace Delete	#41	DEL	DEL — delete the character at the cursor
Escape	#42	AC	AC — clear all of the current calculation
4	#43	4	Digit 4
5	#44	5	Digit 5
6	#45	6	Digit 6
* x	#46	×	Multiplication ×
/	#47	÷	Division ÷
1	#48	1	Digit 1
2	#49	2	Digit 2

3	#50	3	Digit 3
+	#51	+	Addition +
-	#52	–	Subtraction –
0	#53	0	Digit 0
.	#54	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#56	E	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
Enter =	#57	EXE	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (6)

Work each one on calculator #43, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. **slope(2,1)** = _____
2. **ode(0,1,1)** = _____
3. **euler(0,1,1,10)** = _____
4. **divg(0,0)** = _____
5. **jdet(1,0)** = _____
6. **eqx(0.8,0.1)** = _____

Set B — Key-finding drills (53)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. **Which key is "y' =" (Edit the equation and system)?** _____
2. **Which key is "WINDOW" (WINDOW)?** _____
3. **Which key is "ZOOM" (ZOOM)?** _____
4. **Which key is "TRACE" (TRACE)?** _____
5. **Which key is "GRAPH" (GRAPH)?** _____
6. **Which key is "SHIFT" (SHIFT / 2nd)?** _____
7. **Which key is "FIELD" (Toggle slope field / phase plane)?** _____
8. **Which key is "METHOD" (Euler → Heun → RK4)?** _____

9. Which key is “▲” (Cursor ▲)? _____

10. Which key is “▼” (Cursor ▼)? _____

11. Which key is “◀” (Cursor ◀)? _____

12. Which key is “▶” (Cursor ▶)? _____

13. Which key is “OK” (OK)? _____

14. Which key is “TOOLS” (TOOLS)? _____

15. Which key is “ON” (ON)? _____

16. Which key is “x” (Variable x)? _____

17. Which key is “y” (Variable y)? _____

18. Which key is “HOME” (HOME)? _____

19. Which key is “MODE” (SETUP)? _____

20. Which key is “y’ (” (Slope field value)? _____

21. Which key is “ODE” (Accurate RK4 solution)? _____

22. Which key is “Euler” (Euler’s method with n steps)? _____

23. Which key is “RK4” (Runge–Kutta 4 with n steps)? _____

24. Which key is “div” (Divergence of the system)? _____

25. Which key is “detJ” (Jacobian determinant)? _____

26. Which key is “eq x” (Equilibrium x from a guess)? _____

27. Which key is “sin” (Sine)? _____

28. Which key is “cos” (Cosine)? _____

29. Which key is “ex” (e^x)? _____

30. Which key is “^” (Power x^y / ^)? _____

31. Which key is “x^2” (Square x^2)? _____

32. Which key is “(” (Opening bracket)? _____

33. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

34. Which key is “,” (Comma / argument separator)? _____

35. Which key is “π” (π)? _____

36. Which key is “SEL” (sel)? _____

37. Which key is “|x|” (Absolute value |x|)? _____

38. Which key is “DEL” (DEL)? _____

39. Which key is “AC” (AC)? _____

40. Which key is “×” (Multiplication ×)? _____

41. Which key is “÷” (Division ÷)? _____

42. Which key is “+” (Addition +)? _____

43. Which key is “−” (Subtraction −)? _____

44. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

45. Which key is “(−)” (Negative sign)? _____

46. Which key is “E” (×10^x / EXP / EE)? _____

47. Which key is “EXE” (Execute / =)? _____

48. How do you reach “TABLE”? _____

49. How do you reach “Heun”? _____

50. How do you reach “eq y”? _____

51. How do you reach “ln”? _____

52. How do you reach “√”? _____

53. How do you reach “Ans”? _____

Appendix A — Drill answers

Set A

1. **1** (display `1`)
   `#20` y′ (→ `#49` 2 → `#34` , → `#48` 1 → `#33`) → `#57` EXE

2. **0.73575888** (display `0.7357588824`)
   `#21` ODE → `#53` 0 → `#34` , → `#48` 1 → `#34` , → `#48` 1 → `#33`) → `#57` EXE

3. **0.69735688** (display `0.6973568802`)
   `#22` Euler → `#53` 0 → `#34` , → `#48` 1 → `#34` , → `#48` 1 → `#34` , → `#48` 1 → `#53` 0 → `#33`) → `#57` EXE

4. **-0.3** (display `-0.3`)
   `#24` div → `#53` 0 → `#34` , → `#53` 0 → `#33`) → `#57` EXE

5. **2** (display `2`)
   `#25` detJ → `#48` 1 → `#34` , → `#53` 0 → `#33`) → `#57` EXE

6. **1** (display `1`)
   `#26` eq x → `#53` 0 → `#54` . → `#39` 8 → `#34` , → `#53` 0 → `#54` . → `#48` 1 → `#33`) → `#57` EXE

Set B

1. #1
2. #2
3. #3
4. #4

5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #36
37. #37
38. #41
39. #42
40. #46
41. #47
42. #51
43. #52
44. #54
45. #55
46. #56
47. #57
48. #6 SHIFT then #5 GRAPH
49. #6 SHIFT then #22 Euler
50. #6 SHIFT then #26 eq x
51. #6 SHIFT then #29 ex
52. #6 SHIFT then #31 x^2
53. #6 SHIFT then #55 (-)

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23	E	#56
Absolute value |x|	|x|	#37
AC — clear all of the current calculation	AC	#42
Accurate RK4 solution	ODE	#21
Addition +	+	#51
Ans — recall the last answer into the current expression	Ans	#6 → #55
Closing bracket). Closing brackets at the end of a line may be omitted	)	#33
Comma / argument separator (between function arguments, list items or coordinates)	,	#34
Cosine (current angle unit)	cos	#28
Cursor ▲ — up / previous history line / previous item	▲	#9
Cursor ▶ — move right in the line	▶	#12
Cursor ▼ — down / next history line / next item	▼	#10
Cursor ◀ — move left in the line	◀	#11
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#54
DEL — delete the character at the cursor	DEL	#41
Digit 0	0	#53
Digit 1	1	#48
Digit 2	2	#49
Digit 3	3	#50
Digit 4	4	#43
Digit 5	5	#44
Digit 6	6	#45
Digit 7	7	#38

Digit 8	8	#39
Digit 9	9	#40
Divergence of the system	div	#24
Division ÷	÷	#47
Edit the equation and system	y′ =	#1
Equilibrium x from a guess	eq x	#26
Equilibrium y	eq y	#6 → #26
Euler → Heun → RK4	METHOD	#8
Euler's method with n steps	Euler	#22
e^x — exponential	ex	#29
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	EXE	#57
GRAPH — draw the graph	GRAPH	#5
Heun's method	Heun	#6 → #22
HOME — return to the main calculation screen	HOME	#18
Jacobian determinant	detJ	#25
Multiplication ×	×	#46
Natural logarithm ln	ln	#6 → #29
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#55
OK — confirm the highlighted item	OK	#13
ON — power on	ON	#15
Opening bracket (	(	#32
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	^	#30
Runge-Kutta 4 with n steps	RK4	#23
sel	SEL	#36
SETUP — angle unit, number format, display options	MODE	#19
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#6

Sine (uses the current angle unit)	sin	#27
Slope field value	y′ (	#20
Square root √	√	#6 → #31
Square x^2	x^2	#31
Subtraction –	–	#52
TABLE — table of values	TABLE	#6 → #5
Toggle slope field / phase plane	FIELD	#7
TOOLS — tools menu	TOOLS	#14
TRACE — move along a graph	TRACE	#4
Variable x	x	#16
Variable y	y	#17
WINDOW — set the graph view	WINDOW	#2
ZOOM — zoom presets	ZOOM	#3
π (3.14159265…) as an exact constant	π	#35

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Differential-equation grapher. Slot 1 is dy/dx = f(x,y); slots 2 and 3 define the autonomous system used in phase-plane mode.

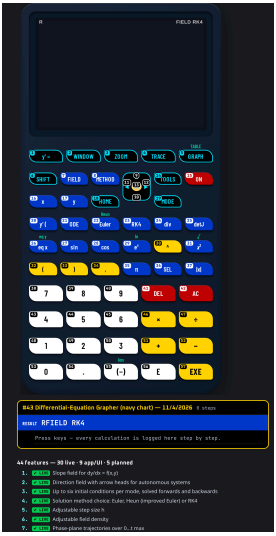
Euler is first order, Heun second order and RK4 fourth order; the comparison tool shows the error against RK4 with 20× smaller steps.

Planned (not yet working)

- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 y′ =	#2 WINDOW	#3 ZOOM
#4 TRACE	#5 GRAPH ⇧ TABLE	#6 SHIFT
#7 FIELD	#8 METHOD	#9 ▲
#10 ▼	#11 ◀	#12 ▶
#13 OK	#14 TOOLS	#15 ON
#16 x	#17 y	#18 HOME
#19 MODE	#20 y′ (	#21 ODE
#22 Euler ⇧ Heun	#23 RK4	#24 div
#25 detJ	#26 eq x ⇧ eq y	#27 sin
#28 cos	#29 ex ⇧ ln	#30 ^
#31 x^2 ⇧ √	#32 (	#33)
#34 ,	#35 π	#36 SEL
#37 |x|	#38 7	#39 8
#40 9	#41 DEL	#42 AC
#43 4	#44 5	#45 6
#46 ×	#47 ÷	#48 1
#49 2	#50 3	#51 +

#52 –	**#53** 0	**#54** .
#55 (–) ⇧ Ans	**#56** E	**#57** EXE

U-45 CALCULATORS · Calculator #43 · cut out and keep